

HAN YI

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🌐 Homepage | 📄 Google Scholar

Summer 2027 research internship availability: May–Aug. 2027

EDUCATION

University of North Carolina at Chapel Hill (UNC)

Ph.D. in Computer Science, GPA: 4.0/4.0 – Expected Graduation: May 2029

Aug. 2024 – Present

Advisor: Prof. Gedas Bertasius

National University of Singapore (NUS)

Master of Computing in Computer Science

Aug. 2022 – Jun. 2024

Advisor: Prof. Tat-Seng Chua

East China University of Science and Technology (ECUST)

B.S. in Mathematics and Applied Mathematics, GPA: 3.7/4.0, Top 10%

Sep. 2018 – Jun. 2022

RESEARCH AREAS

Generative Video | Controllable Generation | Multimodal Learning | Video Understanding

RESEARCH EXPERIENCE

Graduate Research Assistant

UNC at Chapel Hill

Aug. 2024 – Present

Advisor: Prof. Gedas Bertasius

SVI-Bench: Strategic Video Intelligence

ECCV 2026

- Co-developed a 9-task framework for strategic video intelligence over 35K hours of broadcast basketball, soccer, and hockey, aligning video with 15M annotated actions, expert commentary, game reports, and structured statistics across perception, reasoning, simulation, and agency.
- Designed and trained the **Strategic Simulation models (T7/T8)** for controllable multi-player video generation: extended Wan 2.1 with a joint conditioning scheme over initial appearance, player-removed backgrounds, multi-agent trajectories, and goal/spatial constraints, and curated 290K motion-conditioned and 74K goal-conditioned training examples. **Outperformed MagicMotion and ATI** — 0.611 vs. 0.544 Video mIoU on soccer (T7), and 50.2% vs. 31.4% / 40.5% goal accuracy (T8).

ExAct: Expert Action Analysis

NeurIPS 2025

- Led a 3,521-example video-language benchmark for expert-level understanding of skilled physical actions across 11 activities and 6 domains, targeting subtle execution details that are missed by conventional coarse action-recognition benchmarks.
- Designed an expert-grounded data pipeline with hard distractor generation, length-controlled candidate filtering, language-only shortcut removal, and domain-expert verification; the strongest evaluated VLM achieved 55.35% versus 82.02% for human experts.

INDUSTRY EXPERIENCE

Research Intern

Tencent Games

Jan. 2024 – Jun. 2024

- Developed a controllable few-view 3D asset generation pipeline using NeRF-based reconstruction from sparse multi-view images, focusing on geometry/appearance consistency and controllability for game-content synthesis.
- Improved reconstruction quality by leveraging cross-view information and multi-view conditioning, strengthening consistency across novel viewpoints.

ADDITIONAL RESEARCH EXPERIENCE

Research Assistant

Dec. 2022 – Sep. 2023

NExT++ Research Center, National University of Singapore

Advisor: Prof. Tat-Seng Chua

- Proposed a **Multi-Scale Triplane Network (MTN)** for progressive text-to-3D generation and a coarse-to-fine learning strategy that jointly schedules camera radius and diffusion timestep, improving optimization stability and high-resolution detail. Published in ACM TOMM 2026.

Research Assistant

Nov. 2021 – Oct. 2022

Wuhan University

Advisor: Prof. Zhi Gao

- Developed a motion deblurring framework that jointly models image blurring and deblurring; introduced Rear-Blur-COCOMini to reduce the synthetic-to-real blur gap and achieved state-of-the-art performance across multiple benchmarks. Published in IEEE TIP 2023.
- Developed a recurrent single-image raindrop removal network using luminance priors and contextual feature aggregation, with a two-stage detection/removal design that improved generalization on synthetic and real imagery. Published in Neurocomputing 2024.

Research Assistant

Jul. 2021 – Sep. 2021

Tsinghua University

Advisor: Prof. Hao Zhang

- Built a vision-based indoor localization system using Manchester-coded ceiling LEDs and OpenCV SolvePnP; designed calibration for challenging camera geometries and real-world optical distortions.

SELECTED PUBLICATIONS

1. SVI-Bench: A Dynamic Microworld for Strategic Video Intelligence

Yulu Pan*, **Han Yi***, Seongsu Ha*, Md Mohaiminul Islam*, Benjamin Zhang, Lorenzo Torresani, Gedas Bertasius
ECCV 2026 – Paper

2. ExAct: A Video-Language Benchmark for Expert Action Analysis

Han Yi, Yulu Pan, Feihong He, Xinyu Liu, Benjamin Zhang, Oluwatuminu Oguntola, Gedas Bertasius
NeurIPS 2025 – Paper

3. Progressive Text-to-3D Generation for Automatic 3D Prototyping

Han Yi, Zhedong Zheng, Xiangyu Xu, Tat-Seng Chua
ACM TOMM 2026 – Paper

4. Image Deblurring with Image Blurring

Ziyao Li, Zhi Gao, **Han Yi**, Yu Fu, Boan Chen
IEEE Transactions on Image Processing (TIP), 2023

5. Single Image Deraindrop Leveraging Luminance Priors and Context Aggregation

Yi Liu, Zhi Gao, Tiancan Mei, **Han Yi**
Neurocomputing, 2024

SERVICE & AWARDS

Reviewer: NeurIPS 2026; ICLR 2025–2026; ACM Multimedia 2024–2025; ACM Multimedia Asia 2025

Outstanding Reviewer: ACM Multimedia 2024

SKILLS

Programming: Python, C/C++, Bash

Frameworks: PyTorch, TensorFlow, OpenCV, Scikit-learn

Tools: Docker, Git, Linux